

Output: output.txt

=== (a) Max Heap insertion ===

```
Insert 45 : [45]
Insert 72 : [72, 45]
Insert 30 : [72, 45, 30]
Insert 90 : [90, 72, 30, 45]
Insert 65 : [90, 72, 30, 45, 65]
Insert 50 : [90, 72, 50, 45, 65, 30]
Insert 85 : [90, 72, 85, 45, 65, 30, 50]
Heap height = 2, comparisons = 9, swaps = 5
```

=== (b) Heap Sort ===

```
Input : [45, 72, 30, 90, 65, 50, 85]
After build-heap : [90, 72, 85, 45, 65, 50, 30]
Extract 90 -> [85, 72, 50, 45, 65, 30, 90]
Extract 85 -> [72, 65, 50, 45, 30, 85, 90]
Extract 72 -> [65, 45, 50, 30, 72, 85, 90]
Extract 65 -> [50, 45, 30, 65, 72, 85, 90]
Extract 50 -> [45, 30, 50, 65, 72, 85, 90]
Extract 45 -> [30, 45, 50, 65, 72, 85, 90]
Sorted: [30, 45, 50, 65, 72, 85, 90]
Heap Sort: comparisons = 21, swaps = 18
```

=== (b) Quick Sort ===

```
Input : [45, 72, 30, 90, 65, 50, 85]
pivot=85 (range 0..6) -> [45, 72, 30, 65, 50, 85, 90]
pivot=50 (range 0..4) -> [45, 30, 50, 65, 72, 85, 90]
pivot=30 (range 0..1) -> [30, 45, 50, 65, 72, 85, 90]
pivot=72 (range 3..4) -> [30, 45, 50, 65, 72, 85, 90]
Sorted: [30, 45, 50, 65, 72, 85, 90]
Quick Sort: comparisons = 12, swaps = 6
```